of the mean depth of the cargo hold; and

.2 be fitted at the aft e nd of the hold, or above it s lowest part where the inner bottom is not parallel to the designed waterline. Where webs or partial watertight bulkheads are fitted above the inner bottom, Administrations may require the fitting of additional detectors.

4 The water level detectors required by pa ragraph 2 need not be fitted in ships complying with regulation XII/12, or in ships having watertight side compartments each side of the cargo hold length extending vertically at least from inner bottom to freeboard deck."

PART C
MACHINERY INSTALLATIONS

2 The following new regulation 35-1 is in serted after existing regulation 35:

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SOURCE: MSC.194(80).pdf
side of the cargo hold length extending vertically at least from inner bottom to freeboard deck."

PART C
MACHINERY INSTALLATIONS

2 The following new regulation 35-1 is in serted after existing regulation 35:

"Regulation 35-1
Bilge pumping arrangements

1 This regulation applies to ships c onstructed on or after 1 January 2009.

2 Passenger ships and cargo ships

2.1 An efficient bilge pumping system shall be provided, capable of pumping from and draining any watertight compartment other than a space permanently appropriated for the carriage of fresh water, water ballast, oil fuel or liq uid cargo and for which other efficient means of pumping are provided, unde r all practical conditions. Efficient means shall be provided for draining water from insulated holds.

2.2 Sanitary, ballast and ge neral service pumps may be accepted as independent power bilge pumps if fitted with the necessary connections to the bilge pumping system.

2.3 All bilge pipes used in or under coal bunkers or fuel storage tanks or in boiler or machinery spaces, including spaces in which oi l-settling tanks or oil fuel pumping units

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SOURCE: MSC.194(80).pdf
2.3 All bilge pipes used in or under coal bunkers or fuel storage tanks or in boiler or machinery spaces, including spaces in which oi l-settling tanks or oil fuel pumping units are situated, shall be of steel or other suitable material.

2.4 The arrangement of the bilge and balla st pumping system shall be such as to prevent the possibility of wate r passing from the sea and from water ballast spaces into the cargo and machinery spaces, or from one comp artment to another. Provision shall be made to prevent any deep tank having bilge and ballast connections being inadvertently flooded from the sea when containing cargo, or being discharged through a bilge pump when containing water ballast.

2.5 All distribution boxes and manually operate d valves in connection with the bilge pumping arrangements shall be in positions which are accessible under ordinary circumstances.

2.6 Provision shall be made for the drainage of enclosed cargo spaces situated on the

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SOURCE: MSC.194(80).pdf
pumping arrangements shall be in positions which are accessible under ordinary circumstances.

2.6 Provision shall be made for the drainage of enclosed cargo spaces situated on the bulkhead deck of a passenger ship and on the freeboard deck of a cargo ship, provided that the Administration may permit the means of drainage to be dispensed with in any particular compartment of any ship or class of ship if it is satisfied that by reason of size or internal subdivision of those spaces the safety of the ship is not thereby impaired.

2.6.1 Where the freeboard to the bulkhead deck or the freeboard deck, respectively, is such that the deck edge is immersed when th e ship heels more than 5°, the drainage shall be by means of a sufficient number of scupper s of suitable size di scharging directly overboard, fitted in accordance with the requirements of regulation 15 in the case of a passenger ship and the requirements for sc uppers, inlets and discharges of the International Convention on Load Lines in force in the case of a cargo ship.

2.6.2 Where the freeboard is such that the e dge of the bulkhead deck or the edge of the

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SOURCE: MSC.194(80).pdf
International Convention on Load Lines in force in the case of a cargo ship.

2.6.2 Where the freeboard is such that the e dge of the bulkhead deck or the edge of the freeboard deck, respectively, is immersed when the ship heels 5° or less, the drainage of the enclosed cargo spaces on the bulkhead deck or on the freeboard deck, respectively, shall be led to a suitable space, or spaces, of adequate capacity, having a high water level alarm and provided with suitable arrangement s for discharge overboard. In addition it shall be ensured that:

.1 the number, size and disposition of the scuppers are such as to prevent unreasonable accumulation of free water;

.2 the pumping arrangements required by this regulation for passenger ships or cargo ships, as applicable, take account of the requirements for any fixed pressure water-spraying fire extinguishing system;

.3 water contaminated with petrol or other dangerous substances is not drained to machinery spaces or other spaces where sources of ignition may be present; and

.4 where the enclosed cargo sp ace is protected by a carbon dioxide fire extinguishing system the deck scuppers are fitted with means to

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SOURCE: MSC.194(80).pdf
drained to machinery spaces or other spaces where sources of ignition may be present; and

.4 where the enclosed cargo sp ace is protected by a carbon dioxide fire extinguishing system the deck scuppers are fitted with means to prevent the escape of the smothering gas.

3 Passenger ships

3.1 The bilge pumping system required by para graph 2.1 shall be capable of operation under all practicable conditions af ter a casualty whether the ship is upright or listed. For this purpose wing suctions shall generally be fitted except in narrow compartments at the end of the ship where one suction may be su fficient. In compartm ents of unusual form, additional suctions may be re quired. Arrangements shall be made whereby water in the compartment may find its way to the suction pipes. Where, for particular compartments,

the Administration is satisfied that the provisi on of drainage may be undesirable, it may allow such provision to be dispensed with if calculations made in accordance with the

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SOURCE: MSC.194(80).pdf
the Administration is satisfied that the provisi on of drainage may be undesirable, it may allow such provision to be dispensed with if calculations made in accordance with the conditions laid down in regulations 7 and 8 show that the survival capability of the ship will not be impaired.

3.2 At least three power pumps shall be fitted connected to the bilge main, one of which may be driven by the propulsion mach inery. Where the bilge pump numeral is 30 or more, one additional independent power pump shall be provided.

The bilge pump numeral shall be calculated as follows:
when P1 is greater than P: bilge pump numeral = ++ P P VP M1 1272
in other cases: bilge pump numeral =

+ V P M272
where:

L = the length of the ship (metres), as defined in regulation 2;

M = the volume of the machinery space (cubic metres), as defined in regulation 2, that is below the bulkhead deck; with the addition thereto of the volume of any permanent oil fuel bunkers which may be situated above the inner bottom and forward of, or abaft, the machinery space;

P = the whole volume of the passenger and crew spaces below the bulkhead

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SOURCE: MSC.194(80).pdf
the volume of any permanent oil fuel bunkers which may be situated above the inner bottom and forward of, or abaft, the machinery space;

P = the whole volume of the passenger and crew spaces below the bulkhead
deck (cubic metres), which are provided for the accommodation and use of passengers and crew, excluding baggage, store, provision and mail rooms;

V = the whole volume of the ship below the bulkhead deck (cubic metres);